JONATHAN HAAB

Computational Biology and Bioinformatics MSc student

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EDUCATION

Computational Biology and Bioinformatics Master

ETH Zürich

September 2019 - Today Zürich (CH)

I developed my Computer Science skills and got introduced to the field of Artificial Intelligence while deepening my knowledge in Quantitative Biology. I applied what I learned on real-world problems during two short research projects.

Life Sciences Engineering Bachelor

EPF Lausanne

September 2016 – July 2019 Lausanne (CH)

I acquired strong engineering and analytical skills. I made my way through calculus, I had my first encounter with object-oriented programming and I took active part in Life Sciences oriented laboratory experiments.

Electrical Engineering (exchange year program)

Lund University - Faculty of Engineering (LTH)

August 2018 – June 2019 Lund (SE)

I had the chance to discover a new country, improve my English skills and grow my social network. I took part in a wide range of lectures to complete my Bachelor, from Signal processing to Biomechanics.

PROJECTS

Evaluation of computational pipelines for multimodal single-cell data

Robinson Lab DMLS - UZH

May - July 2021 Zürich (CH)

I investigated the performance of various normalization and clustering methods on CITE and REAP-seq data. I developed my own method to score the normalization based on its aptitude to differentiate signal from negative (isotype) and positive control ADTs.

Surrogate data generation using Deep Knockoffs for nonparametric testing of fMRI time courses

MIP:Lab CIBM - EPF Lausanne

February - April 2021 Lausanne/Geneva (CH)

I investigated the novel application of Knockoff filter for fMRI data analysis under the supervision of Ph.D. M. G. Preti. I got my first experience of working in a research group and I improve my knowledge of Python.

VOLUNTEERING

CCBB President

VIS - ETH Zürich

Februar 2021 - Today Zürich (CH)

As a committee, we organised social and career events to improve the life of students and help them benefit from the experience of others. Personally, I learned to lead and manage a small group of people, communicate with them as well as with other committees. I gained in organizational and budgeting skills.

LANGUAGES

French ●●●●●

English ●●●●○

German ●●○○○

TECHNICAL SKILLS

Languages
C++, Python, R, Matlab

Tools
PyCharm, Jupyter, RStudio, GitHub

Document preparation
LaTeX, Microsof Office Suite

SOFT SKILLS

Problem solving | Hard-working | Eye for detail | Dynamic | Initiative taker | Punctual | Respectful | Team worker | Open to discussion

ACHIEVEMENTS

Being elected as committee president
I was accepted as the student association committee president of my Master (2021)

Being selected to study abroad
I was given the opportunity to take part in an exchange program based on my grades and motivation (2018)

Receiving a decoration from the army
I was awarded the Military Sport 1 decoration by the Swiss army during recruitment (2015)

Shooting a documentary
A fellow student and myself presented our movie at the "Visions du Réel" festival in Nyon (2015)

Performing in a painting show
I organised a painting show first alone (2013) and then leading three painters (2014) at the "Caribana" music festival